

Task 2: Query Decomposition

Question 1: List all products

Intent: Retrieve all records from products table

Tables: products

Columns: *

Filters: None

Joins: None

Question 2: Get all customers

Intent: Retrieve all records from customers table

Tables: customers

Columns: *

Filters: None

Joins: None

Question 3: Show all orders

Intent: Retrieve all records from orders table

Tables: orders

Columns: *

Filters: None

Joins: None

Question 4: List all employees

Intent: Retrieve all records from employees table

Tables: employees

Columns: *

Filters: None

Joins: None

Question 5: Get all offices

Intent: Retrieve all records from offices table

Tables: offices

Columns: *

Filters: None

Joins: None

Question 6: Show all product lines

Intent: Retrieve all records from productlines table

Tables: productlines

Columns: *

Filters: None

Joins: None

Question 7: List all payments

Intent: Retrieve all records from payments table

Tables: payments

Columns: *

Filters: None

Joins: None

Question 8: Get product names and prices

Intent: Retrieve specific details for all products

Tables: products

Columns: "productName", "buyPrice", "MSRP"

Filters: None

Joins: None

Question 9: Get customer names and cities

Intent: Retrieve names and locations of customers

Tables: customers

Columns: "customerName", city

Filters: None

Joins: None

Question 10: List employee first and last names

Intent: Retrieve employee names

Tables: employees

Columns: "firstName", "lastName"

Filters: None

Joins: None

Question 11: Get all order dates

Intent: Retrieve dates of all placed orders

Tables: orders

Columns: "orderDate"

Filters: None

Joins: None

Question 12: Show product vendor list

Intent: Retrieve a unique list of vendors

Tables: products

Columns: DISTINCT "productVendor"

Filters: None

Joins: None

Question 13: Get all product codes

Intent: Retrieve identifiers for all products

Tables: products

Columns: "productCode"

Filters: None

Joins: None

Question 14: List all countries from offices

Intent: Retrieve unique countries where offices exist

Tables: offices

Columns: DISTINCT country

Filters: None

Joins: None

Question 15: Show all order statuses

Intent: Retrieve unique status types for orders

Tables: orders

Columns: DISTINCT status

Filters: None

Joins: None

Question 16: Get all payment amounts

Intent: Retrieve the monetary value of all payments

Tables: payments

Columns: amount

Filters: None

Joins: None

Question 17: List all job titles

Intent: Retrieve unique job titles

Tables: employees

Columns: DISTINCT "jobTitle"

Filters: None

Joins: None

Question 18: Get customer phone numbers

Intent: Retrieve contact numbers for customers

Tables: customers

Columns: "customerName", phone

Filters: None

Joins: None

Question 19: Show product MSRP values

Intent: Retrieve MSRP for all products

Tables: products

Columns: "productName", "MSRP"

Filters: None

Joins: None

Question 20: List order numbers

Intent: Retrieve identifiers for all orders

Tables: orders

Columns: "orderNumber"

Filters: None

Joins: None

Question 21: Get orders with customer names

Intent: Link order records to customers

Tables: orders, customers

Columns: "orderNumber", "customerName"

Filters: None

Joins: orders."customerNumber" = customers."customerNumber"

Question 22: Get employees with office city

Intent: Link employees to physical office locations

Tables: employees, offices

Columns: "firstName", "lastName", city

Filters: None

Joins: employees."officeCode" = offices."officeCode"

Question 23: Get payments with customer names

Intent: Link payment records to customers

Tables: payments, customers

Columns: "paymentDate", amount, "customerName"

Filters: None

Joins: payments."customerNumber" = customers."customerNumber"

Question 24: Get order details with product names

Intent: Link line items to product names

Tables: orderdetails, products

Columns: "orderNumber", "productName", "quantityOrdered", "priceEach"

Filters: None

Joins: orderdetails."productCode" = products."productCode"

Question 25: Get products with product line description

Intent: Link products to category descriptions

Tables: products, productlines

Columns: "productName", "textDescription"

Filters: None

Joins: products."productLine" = productlines."productLine"

Question 26: Get customers with sales rep names

Intent: Link customers to employee reps

Tables: customers, employees

Columns: "customerName", "firstName" (as repFirstName), "lastName" (as repLastName)

Filters: None

Joins: customers."salesRepEmployeeNumber" = employees."employeeNumber"

Question 27: Get orders with customer city

Intent: Link orders to purchasing customer city

Tables: orders, customers

Columns: "orderNumber", city

Filters: None

Joins: orders."customerNumber" = customers."customerNumber"

Question 28: Get employees and their manager

Intent: Link employees to their managers (Self-Join)

Tables: employees

Columns: "firstName" (Employee), "firstName" (Manager)

Filters: None

Joins: employees."reportsTo" = employees."employeeNumber"

Question 29: Get orderdetails with product vendor

Intent: Link line items to product vendor

Tables: orderdetails, products

Columns: "orderNumber", "productVendor"

Filters: None

Joins: orderdetails."productCode" = products."productCode"

Question 30: Get payments with customer country

Intent: Link payments to customer country

Tables: payments, customers

Columns: amount, country

Filters: None

Joins: payments."customerNumber" = customers."customerNumber"

Question 31: Count customers per country

Intent: Group customers by country

Tables: customers

Columns: country, COUNT(*)

Filters: None

Joins: None

Question 32: Total payments per customer

Intent: Group payments by customer and sum

Tables: payments

Columns: "customerNumber", SUM(amount)

Filters: None

Joins: None

Question 33: Number of orders per status

Intent: Group orders by status and count

Tables: orders

Columns: status, COUNT(*)

Filters: None

Joins: None

Question 34: Products per product line

Intent: Group products by line and count

Tables: products

Columns: "productLine", COUNT(*)

Filters: None

Joins: None

Question 35: Employees per office

Intent: Group employees by office code and count

Tables: employees

Columns: "officeCode", COUNT(*)

Filters: None

Joins: None

Question 36: Total stock per product vendor

Intent: Group products by vendor and sum stock

Tables: products

Columns: "productVendor", SUM("quantityInStock")

Filters: None

Joins: None

Question 37: Average buy price per product line

Intent: Group products by line and average the buy price

Tables: products

Columns: "productLine", AVG("buyPrice")

Filters: None

Joins: None

Question 38: Orders per customer

Intent: Group orders by customer and count

Tables: orders

Columns: "customerNumber", COUNT(*)
Filters: None
Joins: None

Question 39: Max MSRP per product line

Intent: Group products by line and find highest MSRP
Tables: products
Columns: "productLine", MAX("MSRP")
Filters: None
Joins: None

Question 40: Min buy price per vendor

Intent: Group products by vendor and find lowest buy price
Tables: products
Columns: "productVendor", MIN("buyPrice")
Filters: None
Joins: None

Question 41: Total number of customers

Intent: Count all rows in customers table
Tables: customers
Columns: COUNT(*)
Filters: None
Joins: None

Question 42: Total number of products

Intent: Count all rows in products table
Tables: products
Columns: COUNT(*)

Filters: None
Joins: None

Question 43: Total revenue from payments

Intent: Sum all payment amounts
Tables: payments
Columns: SUM(amount)
Filters: None
Joins: None

Question 44: Average product price

Intent: Calculate average buy price across all products
Tables: products
Columns: AVG("buyPrice")
Filters: None
Joins: None

Question 45: Max payment amount

Intent: Find highest single payment amount
Tables: payments
Columns: MAX(amount)
Filters: None
Joins: None

Question 46: Min payment amount

Intent: Find lowest single payment amount
Tables: payments
Columns: MIN(amount)
Filters: None
Joins: None

Question 47: Count total orders

Intent: Count all rows in orders table

Tables: orders

Columns: COUNT(*)

Filters: None

Joins: None

Question 48: Total quantity in stock

Intent: Sum the stock of all products

Tables: products

Columns: SUM("quantityInStock")

Filters: None

Joins: None

Question 49: Average MSRP

Intent: Calculate average MSRP across all products

Tables: products

Columns: AVG("MSRP")

Filters: None

Joins: None

Question 50: Number of employees

Intent: Count all rows in employees table

Tables: employees

Columns: COUNT(*)

Filters: None

Joins: None